Problem I

Problem (Problem I.6)**:** For $x, m, \omega \in \mathbb{R}^d$ and $0 < \sigma < \infty$, let

$$G_{m,\omega,\sigma} = \left(\sigma\sqrt{2\pi}\right)^{-d} e^{-\frac{(x-m)^2}{2\sigma^2} + i\omega x}.$$

Show that

$$\mathcal{F}[G_{m,\omega,\sigma}] = (2\pi)^{d/2}\sigma^{-d} e^{i\omega m} G_{\omega,-m,\sigma^{-1}}.$$

Solution: Similar to the case of the Gaussian, we may reduce to the case of $d = 1$ using Fubini's Theorem. Therefore, we will write

$$f(x) = \left(\sigma\sqrt{2\pi}\right)^{-1} e^{-\frac{(x-m)^2}{2\sigma^2}} e^{i\omega x} e^{-ix\xi}.$$

Taking $u = x - m$, it follows that

$$\begin{aligned}
\int f(x)\,dx &= \left(\sigma\sqrt{2\pi}\right)^{-1} e^{i\omega m - im\xi} \int e^{-\frac{u^2}{2\sigma^2} + i\omega u - iu\xi}\,du \\
&= \left(\sigma\sqrt{2\pi}\right)^{-1} \exp\left(i\omega m - im\xi - \frac{1}{2}\sigma^2(\omega-\xi)^2\right) \int e^{-\frac{1}{2\sigma^2}\left(u + i\sigma^2(\omega-\xi)\right)^2}\,du \\
&= \left(\sigma\sqrt{2\pi}\right)^{-1} e^{i\omega m} \exp\left(-im\xi - \frac{1}{2}\sigma^2(\omega-\xi)^2\right) \lim_{A\to\infty} \int e^{-\frac{1}{2\sigma^2}\left(u + i\sigma^2(\omega-\xi)\right)^2}\,du.
\end{aligned}$$

Translating $t = u + i\sigma^2(\omega - \xi)$, it can be shown that by using the residue theorem, the translated integral has the same value as the original integral. Thus,

$$\begin{aligned}
&= \left(\sigma\sqrt{2\pi}\right)^{-1} e^{i\omega m} \exp\left(-im\xi - \frac{1}{2}\sigma^2(\omega-\xi)^2\right) \int e^{-\frac{1}{2\sigma^2}t^2}\,dt \\
&= e^{i\omega m} e^{-\frac{1}{2}\sigma^2(\xi-\omega)^2 - im\xi} \\
&= e^{i\omega} G_{\omega,-m,\sigma^{-1}}.
\end{aligned}$$

Problem (Problem I.8)**:** Let $f : (0,\infty) \to [0,\infty)$ be a Borel measurable function. For all constants $a, b > 0$, show that we have the equality

$$\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \int_0^\infty f\left(y^2\right) dy.$$

Solution: Let $u = \frac{b}{ax}$, so that we have $ax = \frac{b}{u}$ and $\frac{b}{x} = au$, while $du = -\frac{b}{ax^2}\,dx$, or $-\frac{b}{au^2}\,du = dx$. Therefore,

$$\begin{aligned}
\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx &= -\frac{b}{a}\int_\infty^0 \frac{1}{u^2} f\left(\left(\frac{b}{u} - au\right)^2\right) du \\
&= \frac{b}{a}\int_0^\infty \frac{1}{u^2} f\left(\left(au - \frac{b}{u}\right)^2\right) du.
\end{aligned}$$

In particular, we have

$$\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \frac{1}{2}\int_0^\infty \left(1 + \frac{b}{au^2}\right) f\left(\left(au - \frac{b}{u}\right)^2\right) du.$$

Therefore, taking $y = au - \frac{b}{u}$, we get $dy = a\left(1 + \frac{b}{au^2}\right) du$, whence

$$\frac{1}{2}\int_0^\infty \left(1 + \frac{b}{au^2}\right) f\left(\left(au - \frac{b}{u}\right)^2\right) du = \frac{1}{2a}\int_{-\infty}^\infty f\left(y^2\right) dy$$

$$= \frac{1}{a}\int_0^\infty f\left(y^2\right) dy$$

Problem (Problem I.9)**:** Suppose $f : (0,\infty) \to \mathbb{R}$ can take positive or negative values. Suppose the right-hand side of the earlier expression is a convergent Lebesgue integral. Then, show the left-hand side also converges and the same equality holds.

Solution: Write

$$f_+ = \frac{1}{2}(f + |f|)$$
$$f_- = \frac{1}{2}(f - |f|)$$

for the positive and negative parts of the function f. By the way we define the Lebesgue integral of a real-valued function, it follows that

$$\int f \, d\mu = \int f_+ \, d\mu - \int f_- \, d\mu,$$

and the integral $\int f \, d\mu$ converges if and only if both $\int f_+ \, d\mu$ and $\int f_- \, d\mu$ are finite. Writing for the previous problem then gives

$$\int_0^\infty f\left(y^2\right) dy = \int_0^\infty f_+\left(y^2\right) dy - \int_0^\infty f_-\left(y^2\right) dy.$$

Since both the integrals of $f_+(y^2)$ and $f_-(y^2)$ are finite, it follows that for each one we have

$$\int_0^\infty f_\pm\left(y^2\right) dy = \int_0^\infty f_\pm\left(\left(ax - \frac{b}{x}\right)^2\right) dx,$$

so that

$$\begin{aligned}\int_0^\infty f(y^2)\, dy &= \int_0^\infty f_+\left(\left(ax - \frac{b}{x}\right)^2\right) dx - \int_0^\infty f_-\left(\left(ax - \frac{b}{x}\right)^2\right) dx \\ &= \int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx.\end{aligned}$$

Problem (Problem I.10)**:** Derive the Fourier transform of the Cauchy distribution by using real variable methods.

Solution: Write

$$\frac{1}{x^2+1} = \int_0^\infty e^{-t\left(x^2+1\right)}\, dt.$$

Then, we may write

$$\int_{\mathbb{R}} \frac{e^{-ix\xi}}{\pi(x^2+1)}\, dx = \frac{1}{\pi}\int_{\mathbb{R}}\int_0^\infty e^{-t\left(x^2+1\right)-ix\xi}\, dt\, dx.$$

Since $\left|e^{-t\left(x^2+1\right)-ix\xi}\right| = e^{-t\left(x^2+1\right)}$ is integrable on $[0,\infty)\times\mathbb{R}$, it follows by Fubini–Tonelli that we may switch the order of integration. Therefore, we have

$$\frac{1}{\pi}\int_{\mathbb{R}}\int_0^\infty e^{-tx^2-ix\xi-t}\, dt\, dx = \frac{1}{\pi}\int_0^\infty e^{-t}\int_{\mathbb{R}} e^{-tx^2}e^{-ix\xi}\, dx\, dt.$$

Taking $u = x\sqrt{2t}$ and completing the substitution yields

$$= \frac{1}{\pi}\int_0^\infty \frac{1}{\sqrt{2t}} e^{-t} e^{-\frac{\xi^2}{4t}} \int_{\mathbb{R}} e^{-\frac{1}{2}\left(u + i\frac{\xi}{\sqrt{2t}}\right)^2}\, du\, dt.$$

Taking the transformation on the u-integral as $u \mapsto z - i\frac{\xi}{\sqrt{2t}}$ and observing a vanishing contour integral yields that the u-integral equals $\sqrt{2\pi}$. Thus, we have

$$= \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{1}{\sqrt{t}} e^{-\left(t + \frac{\xi^2}{4t}\right)}\, dt.$$

Substituting $s = \sqrt{t}$ and assuming $\xi > 0$, we get

$$= \frac{1}{\sqrt{\pi}} e^{-\xi} \int_0^\infty e^{-\left(s - \frac{\xi}{2s}\right)^2}\, ds.$$

Using Cauchy's trick then yields

$$= e^{-\xi}.$$

Meanwhile, if $\xi < 0$, we have

$$= \frac{1}{\sqrt{\pi}} e^{\xi} \int_0^\infty e^{-\left(s - \frac{-\xi}{2s}\right)^2}\, dx$$
$$= e^{\xi}.$$

Finally, when $\xi = 0$, we have that the integral is given by

$$= \frac{1}{\sqrt{\pi}} \int e^{-s^2}\, ds$$
$$= e^0.$$

Thus,

$$\int_{\mathbb{R}} \frac{e^{-ix\xi}}{\pi(x^2+1)}\, dx = e^{-|\xi|}.$$